

KG-TABI: Design and Auditability of a Conservative Research-Gap Hypothesis Protocol

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Abstract. Systematic literature reviews can surface research opportunities, but sparse graph structure is not itself evidence of a research gap. We present KG-TABI as an *abstract-level, auditable protocol* that turns graph signals into hypotheses about potential research gaps for expert review. The protocol combines event-aggregated knowledge graphs, seeded Louvain communities, a symmetric cut-edge fraction, an FDR-controlled temporal screen, and Toulmin-Abductive Bucketed Inference (TABI). On a legacy-screened 149-paper microservices-security diagnostic, no community passes the isolation gate and no temporal signal survives FDR; consequently, TABI produces no hypothesis. A provenance-recorded re-run from the overlapping 185-record snapshot again produces no eligible signal. A separate dry run demonstrates packet generation only. Thus, the evidence supports implementation behavior, traceability, and a declared null decision—not candidate novelty, usefulness, source support, generalizability, or superiority over a baseline.

Keywords: Auditable Research Protocol · Potential Research-Gap Hypotheses · Knowledge Graphs · Toulmin Argumentation

1 Introduction

Systematic literature reviews (SLRs) consolidate evidence and identify unresolved questions, but manual review is increasingly difficult at modern publication volumes [9]. Search, screening, and graph-based analysis can support this work, yet an absent graph edge can equally reflect incomplete retrieval, noisy extraction, or entity-resolution error. We therefore distinguish a *signal for inspection* from a validated scientific gap.

KG-TABI is a conservative, traceable protocol for moving from a graph signal to a *hypothesis about a potential research gap*. It builds a temporally annotated graph from literature text, detects structurally isolated communities and declining normalized activity, and asks an LLM to express any surviving signal in a Toulmin-style form. The LLM does not certify novelty, importance, truth,

or feasibility. Those properties require source closure and independent domain review.

Our contributions are:

- a provenance-aware abstract-to-graph pipeline with complete-link entity resolution;
- deterministic structural and FDR-controlled temporal signal gates;
- a Toulmin-structured abductive hypothesis schema that separates graph Grounds, proposed Claim, Warrant, feasibility, and validation status; and
- a historical diagnostic, provider-field-recorded forward reproduction, sanity checks, and blinded-packet dry run.

The full algorithms, sensitivity tables, audit packet specifications, and archived diagnostic details are provided in the supplementary material; this is a protocol and diagnostic paper, not a completed effectiveness evaluation of hypothesis generation.

We ask three deliberately bounded research questions: **RQ1**, does the frozen implementation return the declared structural and temporal decision on its recorded inputs? **RQ2**, how sensitive is that decision to partition seeds, traceability restrictions, synthetic perturbations, and alternative size rules? **RQ3**, does the implementation preserve an auditable chain from a frozen input record to a candidate or an explicit null? Candidate novelty, usefulness, and comparative quality are outside these diagnostic RQs and require the confirmatory study specified in Section 5.

2 Background and Scope

Scientific knowledge graphs represent entities and relations extracted from research text [6,11]. Literature-based discovery and embedding-based science mapping can suggest useful cross-literature connections [16,18], but they do not establish that a proposed connection is absent, novel, or important. Graph-augmented retrieval systems improve grounding [3,4,5]; they do not remove the need for evidence appraisal.

Link-prediction methods, including variational graph auto-encoders, estimate likely unobserved edges from graph structure and features [8]; their usual target is recovery of held-out observed links, not validation of a research gap. Recent LLM work studies open-domain hypothesis discovery, literature–data synthesis, and literature-grounded generation with provenance and human evaluation [20,10,19]. KG-TABI is narrower: it records graph grounds and an explicit warrant for expert inspection, while leaving novelty and scientific value to source closure and blinded review. Its present evidence concerns decision traceability rather than candidate-quality gains.

We use a working taxonomy informed by SLR and systematic-mapping practice [9,13]: Explicit, Contradictory Evidence, Methodological, Context and Population, Coverage Intersection, and Evaluation gaps. It is neither expert-validated nor exhaustive. The current prototype operationalizes only a possible Coverage

Table 1. Scope comparison with cited workflow families. “Not a protocol target” describes the primary scope of the cited work, not an empirical benchmark or a claim that a feature is impossible.

Workflow family	Primary output	Signal gate	Trace contract	Review protocol
Scientific KG [11,6]	IE / scientific representation	repre- not a target	extraction/KG record	not a target
Graph retrieval [3,4,5]	re- grounded text	con- retrieval	retrieved context	not a target
Link prediction [8]	predic- held-out links	graph structural prediction	predicted link	not a target
LLM hypothesis discovery [20,10,19]	hypothesis discovery	task-specific	evidence trace	task-specific human evaluation
KG-TABI	potential gap null	or frozen structural temporal	Grounds + claim + warrant → source	→ closure, packet, key, blind audit

Frozen records → recorded screen → abstract chunks → typed event triples → complete-link resolution
→ structural/temporal gates → null record or TABI draft → source closure → blinded review

Fig. 1. Protocol control flow. Only the null branch was exercised under the frozen primary configuration; candidate closure and review remain a specified hand-off, not a validated result.

Intersection signal; the other categories require additional text, study-design, or contradiction analysis.

Protocol novelty. KG-TABI does not claim a new community algorithm, trend test, or LLM inference theory. Its contribution is the integration contract: a frozen signal gate; a field-level chain from corpus decision to evidence span, extraction call, entity-resolution decision, event, signal, and candidate; an explicit distinction between protocol compliance and scientific validity; and a non-generation outcome when the gate fails. Louvain, FDR, embedding resolution, and Toulmin fields are adaptations of established components, while the content-addressed trace and closure/blinding hand-off are the protocol and engineering contributions.

3 KG-TABI Framework

3.1 Forward pipeline and provenance

Given a domain query, the pipeline retrieves metadata, records one inclusion decision per record, partitions available text, extracts typed triples, resolves entities, detects signals, and generates TABI only for signals that pass frozen gates. A forward run records non-secret configuration, input and output hashes, prompt and call provenance, entity-resolution decisions, and a content-addressed

Table 2. Core audit contract in the main protocol. Every arrow is keyed by an ID or SHA-256; raw source text is retained only where the approved storage policy permits.

Stage	Minimum trace fields and audit question
Corpus and chunk	source ID, inclusion decision/reason, source and chunk hashes, stable chunk ID, section and offsets: “which text entered?”
Extraction and resolution	prompt version/hash, configured and provider model metadata, call/response hashes, quote span, typed-label proposal and complete-link decision: “why this triple/node?”
Signal and TABI	graph/configuration hash, event IDs, partition/temporal statistics, compatibility record, TABI call ID: “why was this candidate or null emitted?”
Closure and review	deterministic query variants, returned source IDs/ranks, blind packet ID, separate unblinding key, reviewer labels: “is the claim scientifically supported?”

artifact manifest. This makes a *future* execution inspectable; it does not repair incomplete provenance in the historical diagnostic. Unless stated otherwise, the historical analyses use the undirected projection, Louvain seed 42 and resolution 1.0, $\alpha = .05$, $\beta = .10$, cutoff 2025, $k = 5$, $m = 4$, $\gamma = .30$, and FDR $q < .05$; their shared topology-configuration identifier is `c869fd8a...640b8e` (full SHA-256 in the supplement and artifact).

Terms. An *inclusion decision* is one recorded retain/exclude outcome under the declared screen; a *covered year* has at least one retained event; and *generic nodes* are the predeclared, non-topic-specific labels excluded from temporal testing. A *source closure* is a documented search for evidence that supports, contradicts, or already addresses a candidate. A *post-cutoff event* has a publication year after the last complete analysis year and is excluded. These definitions apply to forward runs; the legacy selection rule and paper IDs are incomplete.

The machine-readable source-trace chain links each record to its chunk, extraction call, triple, resolution decision, event, signal, TABI output, closure, and blind packet. IDs and hashes join every link. The packet fields and non-secret storage rules are specified in the supplement and repository; compliance with this chain is auditable, but does not establish candidate quality.

Triples use six entity types and ten relation types. The controlled vocabularies are:

Entity: METHOD, DATASET, METRIC, CONCEPT, FINDING, TOOL.
 Relation: USES, IMPROVES, ADDRESSES, EVALUATES_ON, PRODUCES,
 CONTRADICTS, EXTENDS, LACKS, COMBINES, APPLIED_TO.

Repeated relations are retained as temporal event records rather than overwritten. Entity resolution first proposes within-type lexical matches and then semantic matches [14]. For a component, it L_2 -normalizes each member-label embedding, averages *all* member vectors, and normalizes that mean to form the

component descriptor; a short canonical label therefore cannot discard alias information. A descriptor match is only a proposal: a union is accepted only when every cross-component typed-label pair passes the active threshold (complete-link cohesion). Thus a high-similarity $A-B-C$ chain cannot merge A and C if they fail the threshold. The resolver is a transparent decision rule, not a demonstrated measure of merge accuracy; blinded labels remain necessary.

3.2 Signals and hypothesis generation

The topology stage projects the event graph to a simple undirected graph and applies seeded Louvain clustering [2]. The projection deliberately asks a narrow adjacency question (which entity labels co-occur at all), while direction, relation type, multiplicity, evidence, and time remain in the event graph and provenance. It is therefore not a typed/directed community model; whether that loss changes candidate quality requires a future ablation. A community C is eligible only if its size ratio is at least α and its symmetric cut-edge fraction is at most β :

$$R_{\text{cut}}(C) = \frac{|E_{\text{cross}}(C)|}{|E_{\text{internal}}(C)| + |E_{\text{cross}}(C)|}.$$

This heuristic is not conductance, a calibrated test, or evidence of a missing scientific relation. The global $\alpha = .05$ rule was frozen for the historical diagnostic, not calibrated as a universally appropriate size rule; component-relative and absolute alternatives are reported only as exploratory diagnostics in the supplement.

Toy structural interpretation. Suppose a community contains {“runtime policy”, “service mesh”} and another contains {“formal verification”, “configuration analysis”}. If both pass the size and low-cut screens, the structural branch records only that their observed adjacency is sparse. It forwards the *pair* only if their declared label-compatibility gate passes; otherwise no LLM call occurs. Even after forwarding, the resulting Claim is a hypothesis about a possible Coverage Intersection, not evidence that the intersection is unstudied. Adjacent communities, low-compatibility pairs, small communities, contradictions, and method/population gaps are intentionally outside this branch’s coverage.

For temporal screening, let $a_{v,y} = n_{v,y}/N_y$ be the incident relation-event count of node v divided by all retained relation events in covered year y . With R the final two covered years and H all earlier covered years, the annual means are defined as $\bar{a}_{v,R} = \frac{1}{|R|} \sum_{y \in R} a_{v,y}$ and $\bar{a}_{v,H} = \frac{1}{|H|} \sum_{y \in H} a_{v,y}$. The activity decline is then $d_v = \text{clip}(1 - \bar{a}_{v,R}/\bar{a}_{v,H}, 0, 1)$ when $\bar{a}_{v,H} > 0$, where $\text{clip}(x, a, b) = \max(a, \min(x, b))$, and $d_v = 0$ when $\bar{a}_{v,H} = 0$. A node must meet event and multi-year minimums, have negative Sen slope, satisfy $d_v \geq \gamma$, and survive Benjamini–Hochberg FDR control [12, 7, 15, 1]. The primary screen tests declining extracted activity, not topic disappearance; it is susceptible to changing extraction density and corpus composition.

TABI converts a surviving signal into a JSON argument with *Grounds*, *Claim*, *Warrant*, and a feasibility bucket, following Toulmin’s argument layout [17]. The abduction is operationalized by inferring a possible explanation (the claim) for the observed topological isolation or temporal decline (the grounds) under a conditional warrant. Grounds are graph statistics, Claim is a testable hypothesis about a potential research gap, and Warrant is an LLM-proposed rationale. The Bucket labels feasibility only: `near_term_feasible` means the proposed study can plausibly use currently available data, methods, and infrastructure, whereas `long_term_or_speculative` indicates a material dependency on unavailable data, immature methods, or unresolved enabling technology. The label is model-proposed, must be reviewed, and is not a probability that the Claim is true. Generated JSON is rejected unless all four fields are non-empty and the Bucket is one of these two values. For illustration, its schema is:

```
{"Grounds": signal, "Claim": hypothesis,
  "Warrant": rationale, "Bucket": feasibility}
```

This is a field schema, not a candidate from the historical null run. A structural candidate uses a *pair* of already eligible communities. The compatibility filter compares their top-degree labels (up to 12), computes lexical overlap and optional embedding similarity, and uses structural isolation only as a ranking feature; it forwards a pair only when the declared content threshold passes. It neither proves a meaningful connection nor creates evidence for the Claim. Every candidate must then undergo source closure and blinded expert assessment; no LLM judge is used as evidence of gap quality.

4 Historical Diagnostic

4.1 Design boundary

We analyze a microservices-security artifact created from a frozen snapshot of 314 retrieved abstract records. The snapshot preserves record identifiers and content but not a replayable account of the original live query expansion and API response; the retrieval itself is therefore not reproduced by this diagnostic. Its 149 retained papers are *legacy-screened*: the stored score and reason fields do not identify a replayable inclusion rule, and the original model, prompt, and call history of its 800 triples are unrecoverable. We therefore do not claim end-to-end reproducibility, screening accuracy, triple accuracy, or entity-merge accuracy for this diagnostic.

To provide a reproducible input for a forward run, the current lexical rule was replayed over all 314 records. It retains 185 records, with 140 overlapping the historical 149. The snapshot records retained metadata, abstract chunks, every inclusion decision, and source/rule hashes, but it is not a replacement result or an external replication: it shares the domain, most records, abstract-only representation, graph construction assumptions, and frozen gates. Detailed comparative counts are in the supplement.

4.2 Null result and diagnostic checks

With $\alpha = .05$, $\beta = .10$, Louvain seed 42, and resolution 1.0, no community passes the isolation gate; the lowest size-eligible R_{cut} is .160714. The temporal screen tests six of 831 non-generic nodes and yields no FDR signal. Across 1,200 partitions and a quote-routed 774-triple restriction, the same primary null is retained. Synthetic injection detects 33/40 linear and 32/40 exponential sustained declines at 75% nominal reduction (95% Wilson intervals in the supplement), while stable/increasing controls are 0/40. Boundary masking conditionally recovers 73/75 induced targets at 75% masking. These are implementation checks, not evidence of novelty, scientific usefulness, or source support.

5 Evaluation Path and Limitations

5.1 Provider-field-recorded forward rerun

The historical null result cannot validate the protocol: it may be caused by corpus selection, abstract-only coverage, extraction error, entity resolution, or conservative gates. We therefore executed a forward end-to-end run (ID **e2e-v1**) from the frozen lexical input snapshot (185 records) with every provider-exposed field in the declared trace contract recorded. The run extracted 1,018 triples, resolved entities using the complete-link resolver with **all-MiniLM-L6-v2** embeddings, and constructed a graph of 1,149 nodes and 1,011 edges. Under the frozen global gate, the largest community has 49 nodes (4.2%), below the implied minimum of 58, so zero communities pass. The graph has 183 connected components and 201 Louvain communities; this is a reproducibility check of the declared null decision, not evidence of generalizability or an independent replication.

The gate diagnostic makes the dependence on this design explicit. If the same bridge rule is retained but the size rule is changed only for exploration, an absolute 10-node rule passes 6 communities and a component-relative 5%-with-10-node-floor rule passes 5; neither profile was used to generate TABI candidates or selected as a replacement primary configuration. The forward artifact has complete paper IDs, enabling a separate paper-normalized sensitivity: 12 nodes meet the five-unique-paper minimum, 6 are tested across four covered years, and 0 survive the unchanged temporal gates. This agrees with the forward temporal null but does not validate the event-normalized historical trend, for which paper IDs are incomplete. Although the forward run records the declared provider-exposed fields, its screening, extraction, and merge outputs remain unvalidated by human labels; reproducibility therefore does not imply graph fidelity.

5.2 Provider-identity audit

All 185 extraction calls requested one configured model string, succeeded on their first attempt, and retained a prompt hash/version, provider response ID, returned model identifier, response/content hashes, and extraction call ID. The endpoint nevertheless returned five identifiers, with 61, 60, 23, 21, and 20 calls

respectively; their descriptive mean triple counts per call range from 4.85 to 6.65 (full labels and counts appear in the supplement). This is not a model comparison: abstracts differ across calls and no input was repeated across routes. It establishes completeness only to provider-exposed metadata, not that a rerun can reproduce underlying routing, weights, or outputs. The machine-readable audit and its input hashes are released with the run.

5.3 Exploratory TABI and blinding dry run

To verify the mechanics of the blinded evaluation toolchain, we replayed the implementation on a legacy pilot corpus (**DefineGap**). The **DefineGap** corpus contains 100 papers on literature review methodologies (abstracts only, screened via a 0.7 relevance filter), resulting in a graph of 550 nodes and 425 edges. In the default configuration (size gate ≥ 28 nodes), 2 communities passed the structural gate, but 0 temporal signals survived. For the relaxed exploratory run, we set event minimum $k = 3$ (from 5), year minimum $m = 3$ (from 4), decline threshold $\gamma = 0.20$ (from 0.30), disabled the Sen slope requirement, and bypassed FDR control. This exploratory configuration identified 10 relaxed-gate temporal signals, generating 10 TABI candidates.

To stress-test packet generation, the implementation shuffled these 10 temporal TABI items with 300 baseline-format items generated by a direct-LLM baseline. The two structural communities do not constitute a reported structural candidate result; this dry run records no evaluated structural-pair outcome. This 310-item artifact was a software dry run and is not the planned balanced evaluation sample. For an actual evaluation study, a balanced candidate budget must be used to prevent baseline dominance and reviewer fatigue. Furthermore, the **DefineGap** items were generated only after the confirmatory protocol had been specified and were used solely to test packet construction; no ratings were collected, so the paper makes no candidate-quality claim. An abbreviated illustrative pilot candidate is provided below. It is a human-readable summary rather than the full record-to-call trace; a confirmatory candidate must additionally expose its paper, chunk, event, configuration, and call identifiers through the released packet:

Illustrative DefineGap candidate (not expert-validated):

Signal ID: DefineGap-temporal-scoping-review

Grounds: In the 100-abstract DefineGap pilot, concept ‘scoping review’ has $n_H = 12$ incident extracted events in the historical window and $n_R = 0$ events in the recent window. Under the exploratory configuration ($k = 3, m = 3, \gamma = 0.20$), its normalized decline score is $d_v = 1.00$, and raw $p = 0.0012$. This signal may reflect corpus coverage or extraction noise and does not establish that research activity has disappeared in the wider literature.

Claim: To what extent can LLM-assisted living scoping-review pipelines continuously update software-engineering evidence maps without reducing screening recall or reviewer agreement?

Warrant: If recurring literature search and screening can be automated while retaining expert verification, living scoping reviews may reduce manual update effort. This expected benefit warrants empirical evaluation, but the present graph signal does not establish pipeline effectiveness.

Warrant Status: llm_proposed_unverified

Closure Status: not run

5.4 Planned candidate evaluation

A future confirmatory study must freeze the corpus rule, source and configuration hashes, model identifiers, prompts, seeds, candidate budget, closure procedure, outcomes, and analysis plan before candidate generation. It must use source closure and at least three independent reviewers who are blinded to system identity and receive the same source bundle for every candidate. They should rate source support, novelty after closure, importance, actionability, feasibility, whether the claim is already addressed, and potential hallucination. The comparison must include a direct-LLM baseline with matched corpus, source bundle, schema, length, and candidate budget. A graph-ground ablation must compare full Grounds against concept-only, abstract-bundle-only, and shuffled-Grounds conditions. To protect blinding integrity, all compared systems must output using the identical Toulmin schema format (Grounds, Claim, Warrant, Bucket), with standardized length constraints and formatting. After rating, reviewers should be asked to guess the generating system’s identity; blinding is considered intact if the guessing accuracy (A_{guess}) does not exceed chance. While a budget of 10 candidates per system is suitable for a formative pilot (reporting medians, raw ratings, agreement, and uncertainty without significance claims), a confirmatory study must specify a larger balanced sample (e.g., 30–50 candidates per system) calibrated via a pre-specified power calculation and repeated in at least two independent domains.

5.5 Limitations

The largest present limitations are clear. The diagnostic is a single small, legacy-screened, abstract-only domain; it has no human labels for screening, extraction, or resolution; its temporal analysis has six eligible nodes; and its perturbation targets are derived from the same graph. The global relative-size gate can suppress meaningful communities in a fragmented graph and is not independently calibrated. Temporal activity is event-normalized rather than paper-normalized in the historical artifact, so it can conflate topic activity with extraction density and is not a paper-normalized trend estimate. The configured extraction endpoint returned multiple provider model identifiers during the forward run, despite a single configured model string; this is recorded in the call log and further limits model-stability claims. Complete-link entity resolution, FDR control, and provenance reduce avoidable ambiguity but do not establish factual correctness or generalizability.

6 Conclusion

KG-TABI is an auditable protocol for making graph-derived *hypotheses about potential research gaps* inspectable, not for automatically validating gaps. In the historical microservices-security diagnostic, neither topology nor temporal screening passes the frozen gates, so the correct result is no generated hypothesis.

The accompanying robustness, synthetic, masking, gate, and paper-normalized sensitivity checks support limited claims about implementation behavior and configuration dependence only. The provenance-recorded forward rerun repeats the conservative null decision on an overlapping snapshot; it is not an independent replication. This evidence does not establish that TABI produces novel, useful, source-supported, or baseline-superior hypotheses. The next evidential step is a multi-domain, candidate-level source-closure and blinded expert comparison with a direct-LLM baseline and graph-ground ablations.

Availability. Source code, audit packets, configuration, and a content-addressed working-tree manifest are available at [the project repository](#). Root-level legacy outputs are separated from paper-backed run artifacts by the repository scope note. At the time of writing there is no reviewed immutable tag or archival DOI; these are release requirements, not completed claims.

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